Supplementary Information

Ecological assets behind improving accounts: national forest evidence and biomass-change identifiability

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S1 National account coverage and availability criteria

The source workbook is the National Ecological Footprint and Biocapacity Accounts, Edition 2026, results and metadata version 1.0. The `national_data` sheet contains 13652 records, 51 columns and 236 country and territory identifiers. The primary interval comprises nine annual observations, 2015–2023 inclusive. A country identifier satisfies the account availability criteria if:

1. every year in the interval is present with a unique country–year key;
2. each publication flag is `+++`, `++` or `+`;
3. all five biocapacity components, their total, forest area, cropland area and population are finite and nonnegative, with positive population;
4. the absolute difference between the five-component sum and the reported total does not exceed 3 gha.

The 3 gha bound follows from rounding five components and one total to the nearest hectare equivalent: the maximum discrepancy is $5(0.5) + 0.5 = 3$ gha. The largest observed discrepancy is 2 gha. Publication flags differ in trade and consumption detail; the three included flag classes all release the biocapacity and production fields used here. The workbook's actual field is named `flag`, while its definition sheet also uses the term `quality_flag`.

There are 183 identifiers satisfying these criteria and 53 not satisfying them. The complete identifier-level table records missing years, publication scope, required-value availability and component consistency. The selected countries in Table S1 all satisfy the account criteria. Their physical boundary and measurement compatibility are not determined by this result.

Table S1: Countries selected for evaluation of physical observation compatibility. Account availability refers to the complete 2015–2023 interval under the criteria in this section.

Country	ISO identifier	Account availability
Brazil	BRA	Complete
Indonesia	IDN	Complete
Democratic Republic of the Congo	COD	Complete
Cameroon	CMR	Complete
Canada	CAN	Complete
Finland	FIN	Complete
France	FRA	Complete
Australia	AUS	Complete

The selection is purposive rather than probabilistic. The availability result cannot be extrapolated to the prevalence of account–asset agreement or disagreement. The 2024–2025 workbook

records do not form part of the primary interval. Account areas remain inputs to the account observation system rather than independent validation measurements of forest extent.

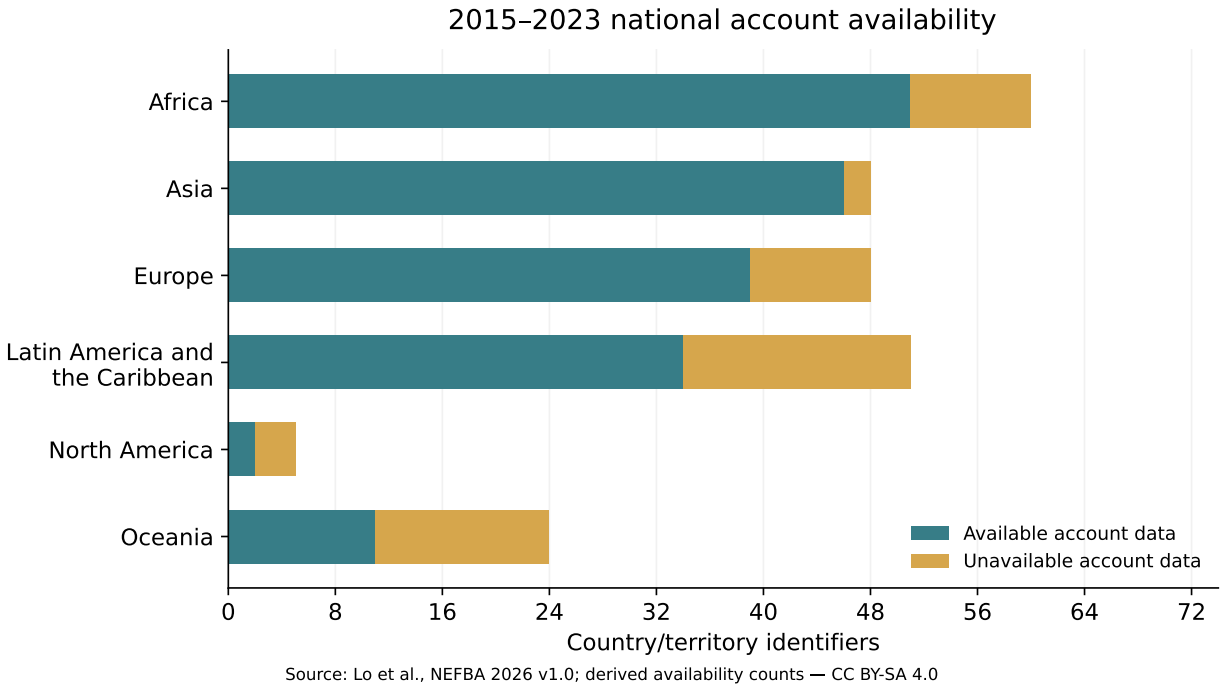


Figure S1: Regional availability of national accounts under the primary eligibility rules. This describes account availability, not physical-data availability or ecological condition. The original separate-row and wrapped-label layout is preserved. Data: Lo et al. (2026); CC BY-SA 4.0.

Table S2: Complete eight-country account-domain sensitivity, 2015–2023 (percentage change). Fishing/built exclusions remain gha quantities; per person is a different denominator. No column is independently observed physical forest change.

ISO	Total	No fishing	No built	Neither	Forest	Per person
BRA	-1.21	-1.24	-1.34	-1.37	-3.15	-5.64
IDN	+5.46	+7.57	+4.64	+6.68	-5.92	-1.82
COD	-2.91	-2.96	-3.36	-3.42	-7.64	-25.63
CMR	+0.54	+0.55	-0.22	-0.24	-3.16	-19.34
CAN	+0.08	-1.41	+0.06	-1.44	-1.10	-8.41
FIN	-1.60	-2.03	-1.54	-1.96	-1.01	-3.74
FRA	-4.47	-4.67	-4.53	-4.75	+2.91	-6.66
AUS	+12.95	+17.30	+12.53	+16.78	-0.33	+2.26

S2 Biomass products, units and spatial support

The source is ESA CCI Biomass v7.0, DOI https://doi.org/10.5285/6429d1aafe1e43b9b414e4a5a7f8b903. Metadata inspection used the 50 km aggregate and the 2010–2020 change product. Their exact names, byte sizes, hashes and retrieval URLs are supplied in the source register. Inspection of these coarse files was limited to coordinates, time values, dimensions and metadata. Separate bounded native-resolution windows at 2015 and 2023 were read to count valid, zero and missing observations; no national biomass stocks or change signs were estimated.

Country–product intersections require a territorial definition, including overseas areas where relevant. Native-resolution or appropriately aggregated forest observations must distinguish no-

Table S3: Observation definitions in the inspected CCI products.

Attribute	Definition or interpretation
Grid	720 longitude by 360 latitude cells, with 0.5° spacing in WGS84 coordinates. The nominal 50 km designation is not an equal-area definition.
Epochs	2005–2012 and 2015–2024, comprising 18 annual epochs. No observed 2013 or 2014 map is supplied in this series.
Biomass density	Above-ground living woody biomass in $\mathrm{Mg\,ha^{-1}}$. The aggregate is normalized by total grid-cell area, not forest area.
Aggregate uncertainty	The guide describes uncertainty in the aggregated mean using variance and spatial covariance contributions. The variable name `agb_sd` alone does not establish an independent-pixel error model.
Change example	Difference, standard deviation and quality-class layers for 2010–2020. This example does not validate change estimates for 2015–2023.
Temporal bounds	The change file's time-bounds attribute names `time_bnds1`; no variable with that name is stored in the file. The documented epoch pair defines the interval considered here.
Observation coverage	The guide specifies a native product domain between 80°N and 60°S, subject to remote-sensing availability. The global coordinate rectangle is not a valid-observation mask.

data cells, observed zero values, nonforest cover and converted land. An estimated density of zero is not, by itself, an independent land-cover observation. Density averaging, mask application and area normalization must refer to the same support. Coarse aggregates do not identify fine-resolution forest-to-cropland transitions.

S3 Change-quality classes and uncertainty

The inspected provider implementation derives quality classes from endpoint biomass estimates and their uncertainty intervals, followed by a growth-limit condition and spatial filtering. The class definitions are summarized in Table S4. These labels describe the provider decision rule; they are not independent observations of ecological change.

Table S4: Provider change-quality classes. Descriptions refer to the processing rule rather than independently validated confidence levels.

Code	Interpretation
0	Both endpoint biomass estimates are zero
1	Strong loss classification
2	Possible loss classification
3	No resolved change under the processing rule
4	Possible gain classification
5	Strong gain classification

The example implementation computes change standard deviation as $\sqrt{\sigma_0^2 + \sigma_1^2}$ and uses one-standard-deviation endpoint intervals in its classification operations. In general,

$$\mathrm{Var}(\widehat{d}_1 - \widehat{d}_0) = \sigma_1^2 + \sigma_0^2 - 2\,\mathrm{Cov}(\widehat{d}_1, \widehat{d}_0). \tag{S1}$$

The variance sum corresponds to zero temporal covariance. A one-standard-deviation interval rule combined with growth and filtering operations does not define a calibrated 95% test. Temporal covariance, systematic biases and the decision rule's operating characteristics require separate evaluation for the specified observation domain.

For an area-weighted total $\widehat{Q} = \sum_i a_i \widehat{d}_i$ on a fixed support, the measurement variance is $\boldsymbol{a}^{\mathsf{T}} \Sigma \boldsymbol{a}$, where Σ is the joint error covariance matrix. Spatial independence reduces this to a

sum of diagonal terms, but that simplification requires evidence. Uncertain support, missing or converted cells, and forest classification error introduce additional terms not represented by a density-only covariance matrix. Product-quality classes cannot provide independent validation when they are constructed from the target maps themselves.

S3.1 Spatial averages, endpoint dependence and systematic error

For a density average $\widehat{\bar{d}} = \sum_i w_i \widehat{d}_i$ with fixed weights satisfying $\sum_i w_i = 1$, the random-error variance is

$$\mathrm{Var}(\widehat{\bar{d}}) = \boldsymbol{w}^{\mathsf{T}} \Sigma \boldsymbol{w}. \tag{S2}$$

For equal weights this is $N^{-2} \sum_i \sum_j \Sigma_{ij}$, reducing to $N^{-2} \sum_i \sigma_i^2$ under independence. An area-weighted mass total instead uses weights a_i in hectares and has variance in Mg^2. Average and total normalization cannot be interchanged. The validation plan gives the equal-weight N^{-2} covariance sum (PVP v6, p. 26).

Write each endpoint density vector as $\widehat{\boldsymbol{d}}_t = \boldsymbol{d}_t + \boldsymbol{b}_t + \boldsymbol{\varepsilon}_t$, where $\boldsymbol{b}_t$ is systematic error and $E(\boldsymbol{\varepsilon}_t) = 0$. On common fixed support, the estimation error of the mass difference is

$$\widehat{\Delta Q} - \Delta Q = \boldsymbol{a}^{\mathsf{T}}(\boldsymbol{b}_1 - \boldsymbol{b}_0) + \boldsymbol{a}^{\mathsf{T}}(\boldsymbol{\varepsilon}_1 - \boldsymbol{\varepsilon}_0). \tag{S3}$$

With $\Sigma_{st} = \mathrm{Cov}(\boldsymbol{\varepsilon}_s, \boldsymbol{\varepsilon}_t)$, its random-error variance, conditional on the support and systematic terms, is

$$\mathrm{Var}(\widehat{\Delta Q}) = \boldsymbol{a}^{\mathsf{T}}(\Sigma_{11} + \Sigma_{00} - \Sigma_{10} - \Sigma_{01})\boldsymbol{a}. \tag{S4}$$

This identity neither removes systematic error nor represents uncertain classification, within-cell allocation, missingness or support changes. The cross-year matrices must be compatible with a valid joint covariance structure. Positive or negative endpoint dependence can respectively reduce or increase random difference variance; neither direction is assigned without evidence.

The uncertainty budget describes within-stack retrieval covariance for C-band and L-band observations (E3UB v7, pp. 14–22), followed by a spatial error-correlation model fitted across 142 one-degree cells from eight regional reference datasets (pp. 23–26). It explicitly notes that these data are not representative of global forests and treats the model as a plausible assumption. The within-stack L-band correlation of 0.5 and the spatial ensemble coefficient are not estimates of the final-map 2015–2023 error covariance for this footprint. Existing propagated components must not be counted twice. The change-variance expression on p. 28 omits cross-year covariance and introduces additional bias-correction uncertainty only if such a correction is applied at the bias-map scale. No local covariance, bias correction or calibrated coverage is inferred from those expressions.

S4 Geographic area calculation

For a rectangle bounded by longitude λ_w, λ_e and latitude ϕ_s, ϕ_n, all expressed in radians, the WGS84 area in hectares is

$$A = \frac{a^2(1-e^2)}{10^4}(\lambda_e - \lambda_w)\left[F(\sin\phi_n) - F(\sin\phi_s)\right], \tag{S5}$$

where $a = 6378137\,\mathrm{m}$, $f = 1/298.257223563$, $e^2 = f(2-f)$ and

$$F(s) = \frac{s}{2(1-e^2s^2)} + \frac{\operatorname{atanh}(es)}{2e}.$$

The formula integrates the ellipsoidal area element between meridians and parallels. Longitude intervals crossing the dateline are split before integration. Geographic cell area varies with latitude even when angular spacing is constant.

For the full ellipsoid, the rectangle integral reduces to

$$A_{\mathrm{WGS84}} = \frac{2\pi a^2}{10^4}\left[1 + \frac{1-e^2}{e}\,\operatorname{atanh}(e)\right].$$

Areas of adjacent latitude bands add exactly at the analytic level because their boundary terms telescope. These identities specify geometric consistency independently of ecological observation accuracy.

S5 Extent, density and spatial compensation

For positive forest areas, expanding the symmetric terms gives

$$\begin{aligned}\frac{(A_1 - A_0)(D_1 + D_0) + (D_1 - D_0)(A_1 + A_0)}{2} \\ = A_1 D_1 - A_0 D_0 = Q_1 - Q_0.\end{aligned} \tag{S6}$$

The identity assigns a pure area change entirely to the extent term and a pure mean-density change entirely to the density term. Mean forest density is undefined at a zero-area endpoint; appearing or disappearing strata therefore require a separately specified allocation convention.

For local net stock changes, $G-L = \sum_i \Delta Q_i$ and $G+L = \sum_i |\Delta Q_i|$. Consequently, $0 \le K \le 1$ when $G + L > 0$. A one-direction change gives $K = 0$, while equal positive and negative sums give $K = 1$. The offset mass is $\min(G, L)$ and has biomass-mass units. When every local change is zero, K is undefined.

These quantities distinguish a national endpoint from its spatial composition, but measurement error can produce the same pattern. If true change is zero and observed local changes are errors e_i, positive and negative errors may yield a large $\sum_i |e_i|$ while $\sum_i e_i$ is close to zero. The resulting high cancellation fraction does not establish ecological redistribution. Interpretation therefore requires an error-model reference at the effective observational support. Resampling to a finer grid does not provide additional independent physical information.

The sums represent local net changes. Gross growth, recruitment, harvest and mortality require additional observations or a separately identified demographic model.

S6 Validation, information dependence and inferential scope

Account availability, physical observation compatibility and validation are distinct requirements. A physical comparison requires stable territorial boundaries, a justified forest support, matched observation dates and a change-error model. Transition validation must distinguish the target forest-to-cropland class from other disturbances and quantify uncertainty using an appropriate reference design. Biomass-level validation does not automatically establish biomass-change precision.

The v7 Product Validation and Intercomparison Report describes independent stock comparisons through 2024 and states that its plot and LiDAR reference data were withheld from production (PVIR v7, pp. 11–14). The reference preparation permits growth-rate adjustment over temporal offsets of up to ten years and applies a forest-fraction correction to small plots (p. 18). Its documented screening includes exclusion of plots deforested between inventory and map dates (p. 13). These are relevant response-design and selection conditions, not direct repeated observations of the present local 2015–2023 forest-stock difference. The reported random-error variance diagnostic omits positional, within-pixel representation and harmonization components; it is not a coverage estimate for a nominal confidence level (p. 19). The report's within-pixel sampling examples show that apparent low- and high-biomass bin biases can arise even when the underlying pixel means are correctly specified (pp. 96–98). Such discrepancies cannot be assigned wholly to map error.

The validation plan identifies repeated NFI and LiDAR observations as potential change references (PVP v6, pp. 17–18, 42), but their dates, uncertainty and availability differ. Its appendix lists a Brazilian LiDAR candidate for 2011–2018 while the accompanying account of the currently retained change-reference data does not include Brazil. Candidate identification therefore does not establish an available, eligible 2015–2023 local validation sample. The change-development plan describes stock differencing, trend methods and further validation needs (ADPC v7, pp. 9–18); a proposed validation activity is not a completed assessment of the present interval. In particular, the trend example's assumption of serially uncorrelated annual estimates cannot substitute for estimation of endpoint covariance.

Practical-change and classification thresholds require calibration independent of held-out evaluation outcomes. The calibration design must evaluate the complete reporting rule under relevant covariance, systematic-bias and missingness scenarios; individual feature tests do not determine the error rate of the combined rule. Comparisons of reporting information require the same cases and weights. Full-population coverage and pairwise overlap must also be reported so that a score change is not attributed to a change in the assessed population. Input lineage must distinguish independent observations from target-derived features and shared calibration/reference data. Algebraically recoverable features do not constitute additional independent measurements.

The local analyses supply account availability, product definitions, support screening, mapped-cover endpoints and conditional biomass-allocation sets. The additional national analyses below supply account trajectories, source-specific mapped extent, endpoint-cohort bounds and conditional inventory-cycle stock quantities. They do not supply calibrated primary national AGB changes, conversion associations, ecological change-resolution probabilities or independent reporting-rule accuracy estimates. Local mapped-cover endpoint areas are reported separately in S8 and do not establish those national outcomes. The additional support observations described in S7 do not establish those outcomes. These quantities cannot be inferred from account publication flags or product access alone.

S7 Territorial and native observation support

Table S5 reports the latitude-domain intersection of the fixed candidate polygons. The geometries are the complete published TopoJSON objects, not the simplified-download objects; their original quantization is retained. Country metadata identifies the source, represented year, licence and full repository commit. Polygon topology was invalid in the released representations for Indonesia, Democratic Republic of the Congo, Canada, Finland and Australia. The polygonal output of a structure-based validity repair is used for geometric screening; repair is not independent validation or a guarantee that omitted islands, water or historical boundary changes are correctly represented.

Table S5: Candidate-boundary domain screening. Areas refer to the selected polygons, not independently measured land or forest area. Full latitude-domain inclusion does not establish valid-data coverage.

ISO	Represented year	Maximum latitude	Outside native domain (%)
BRA	2020	5.27°	0.0000
IDN	2017	5.91°	0.0000
COD	2017	5.39°	0.0000
CMR	2016	13.08°	0.0000
CAN	2020	83.14°	1.2363
FIN	2017	70.09°	0.0000
FRA	2016	51.11°	0.0000
AUS	2021	-9.14°	0.0000

The largest supported polygon by longitude–latitude planar area supplies a representative

interior point for each bounded native observation window. This geometry-only rule is fixed before native-payload retrieval. Each window contains 256 × 256 cells at identical native row and column indices in 2015 and 2023. The native grids have 157,500 rows and 405,000 columns, with approximately 1/1125° spacing; nominal 100 m spacing is not a constant ground area. A pixel-centre country predicate is used only to describe the probe support, not to approximate a national stock integral.

All eight windows contain 65,536 polygon-centre-selected cells with finite, in-range AGB and AGB-standard-deviation values at both endpoints. The declared fill value is −32768; observed zero is retained as a valid value rather than recoded as missing. Zero counts differ across windows and endpoints and are given in Table S6, reproduced from `data/native_probe_coverage_F11.csv`. The identical inherited source is `inherited_F10/data/support/native_probe_coverage.csv`. The supplemental CSV is available at https://github.com/MIKEAA2020/general-sustainability/blob/main/batch%209/agent%202/ecomod_empirical_followup/F11_A01/data/native_probe_coverage_F11.csv. They are not forest-area estimates. These are deterministic local access/support observations, not a probability sample, national coverage assessment, forest classification or validation of biomass change. The exact windows, source responses and counts are identified by query URLs, byte sizes and hashes; raw payloads are not redistributed.

Table S6: Deterministic native access probes, not national or forest-coverage estimates. Every window contains 65,536 cells; joint valid denotes finite, in-range AGB and AGB-SD. Zeros are observed valid AGB values, not missing data or a forest-area classification.

Country identifier	Year	Joint valid cells	AGB zero cells
BRA	2015	65,536	44,546
BRA	2023	65,536	44,767
IDN	2015	65,536	1
IDN	2023	65,536	0
COD	2015	65,536	8,133
COD	2023	65,536	8,167
CMR	2015	65,536	28,273
CMR	2023	65,536	26,975
CAN	2015	65,536	1,206
CAN	2023	65,536	1,072
FIN	2015	65,536	4,226
FIN	2023	65,536	4,328
FRA	2015	65,536	47,398
FRA	2023	65,536	44,301
AUS	2015	65,536	62,358
AUS	2023	65,536	62,479

A separate 1-degree geographic index records polygon intersections for download planning. Multiplying coarse-cell means by these boundary fractions would assume unverified within-cell homogeneity and is not used to estimate biomass. Independent endpoint forest support, land/water treatment, full-support valid-data coverage, uncertainty and territorial correspondence remain required. Unresolved admission is recorded as a hold, not a permanent exclusion or an ecological outcome.

The examined Copernicus land-cover catalogue, form and version constraints list 1992–2022 epochs, with version 2.0.7 for 1992–2015 and version 2.1.1 for 2016–2022. They do not supply the required 2023 forest-support observation. This product-specific endpoint gap does not rule out national or other compatible observations. Neither the primary window nor the conditional secondary window is changed by this finding.

S7.1 Stored uncertainty and zero estimates

The same native payloads contain the AGB and `agb_sd` arrays. Their Int16 units are Mg/ha, declared fill is −32768 and valid range is 0–10,000. Both arrays have valid values at all 65,536 cells per endpoint. The AGB ancillary-variable attribute names `agb_se`, although the listed variable is `agb_sd`; this name discrepancy is not interpreted as data corruption or silently repaired. No separate forest mask, zero-origin field or covariance variable occurs in the examined native variable list.

Table S7: Stored native AGB/SD diagnostics on the fixed Brazilian rectangle. Zero is retained as a valid source value. These counts and quantiles do not measure ecological absence or calibrated uncertainty.

Diagnostic	2015	2023
AGB = 0	44,546	44,767
SD = 0	45,578	45,713
AGB > 0, SD = 0	1,032	946
AGB = 0, SD > 0	0	0
Positive-AGB SD: lower quartile (Mg/ha)	3	3
Positive-AGB SD: median (Mg/ha)	6	7
Positive-AGB SD: upper quartile (Mg/ha)	12	14

Positive AGB at stored SD zero ranges from 1 to 4 Mg/ha at both endpoints. The storage precision alone does not establish whether rounding, clipping, retrieval conventions or another process caused these cases, and no unverified SD floor is imposed. Stored zero/nonzero AGB states contain 44,021 zero–zero, 525 zero–positive, 746 positive–zero and 20,244 positive–positive cells. They exhaust the rectangle, but are not ecological transition classes. The values supply neither final-map cross-year error covariance nor a forest/nonforest allocation of whole-cell biomass.

Table S8: Local mapped-cover endpoints on the identical Brazilian footprint. Values are area-overlap summaries of MapBiomas Collection 11 classifications, not independently validated forest-area estimates. The formations-plus-savanna row overlaps the first row and must not be added to it.

Mapped quantity	2015 (ha)	2023 (ha)
Forest formations	4,319.25	4,510.43
Formations plus savanna	12,169.71	11,435.33
Plantation (separate)	10.66	16.91
Open water (separate)	244.52	320.46

S8 Brazilian mapped-cover fractions and observation operators

The two Collection 11 inputs have the same WGS84 geographic pixel-area grid, with angular spacing 0.000269494585236° and north-west grid origin approximately 74.0209997484°W and 5.4230395387°N. Both are single-band unsigned-byte categorical rasters. The cover window intersects the native biomass indices, rows 105890–106145 and columns 146433–146688. Source generations, ETags, byte ranges and hashes identify the inputs without distributing full country images.

Let C_i be a native biomass cell, L_j a finer land-cover pixel, $\widehat{c}_{jt}$ its mapped class and g a pre-defined group. Its mapped area fraction is

$$\widehat{f}_{ig,t} = \frac{\sum_j A(C_i \cap L_j)\, \mathbf{1}\{\widehat{c}_{jt} \in g\}}{A(C_i)}. \tag{S7}$$

The areas use the WGS84 rectangle integral in S4, including partial boundary pixels. Longitude and latitude integrals factor into separable overlap weights. This is conservative area aggregation, not nearest-neighbour classification, and creates no independent physical information beyond the input classifications. Disjoint group fractions sum to one when the separately retained unobserved group is included. Direct fine-pixel area sums and coarse-fraction-weighted areas agree to less than 3×10^{-8} ha in this footprint; that is an arithmetic closure result, not classification accuracy.

The formations group contains codes 3 (Forest Formation), 5 (Mangrove), 6 (Floodable Forest) and 49 (Wooded Sandbank Vegetation). Savannas 4 and 7 and plantation 9 are separate. The remaining published classes form an explicitly retained other-mapped group, with open water 33 separate. Codes 0 and 27 are reserved for unclassified or legacy unobserved support rather than ecological absence; neither occurs in these bounded cover arrays. An unexpected code is not silently assigned to another class. Individual source-class areas and the alternative formations-plus-savanna and formations-plus-plantation fractions remain available. Adding savannas changes the target cover grouping, not merely its uncertainty.

Published-label completeness is not completeness of direct annual observations. The general Collection 11 ATBD describes temporal gap filling from previous or subsequent classifications, temporal windows, frequency filters and integrated thematic filters. Its seven-year forestry-filter rules for plantation class 9 and biome-specific smoothing can modify timing. Native biomass annual-average epochs and processed annual land-cover labels must therefore not be treated as precisely synchronous instantaneous censuses.

S8.1 Why fine forest area does not identify forest-only biomass

For a mixed native cell, let A_i be full cell area, f_i its true forest fraction and d_i a full-cell biomass density. If d_i^F and d_i^O are the unknown forest and other-land mean densities, then

$$Q_i^{\text{all}} = A_i d_i = A_i\{f_i d_i^F + (1 - f_i) d_i^O\}, \qquad Q_i^F = A_i f_i d_i^F. \tag{S8}$$

For $0 < f_i < 1$, knowing Q_i^{all} and f_i does not in general determine Q_i^F. For example, at $f_i = 1/2$, densities $(d_i^F, d_i^O) = (d_i, d_i)$ and $(3d_i/2, d_i/2)$ give the same full-cell density but forest contributions $A_i d_i/2$ and $3A_i d_i/4$. These are illustrative allocations, not empirical estimates. Nonnegativity alone supplies $0 \le Q_i^F \le Q_i^{\text{all}}$; additional ecological restrictions, finer observations or a validated allocation model would be needed to narrow the compatible allocations. Multiplication by f_i alone imposes equality of the component mean densities, not a consequence of observing forest area. A mapped fraction $\widehat{f}_i$ adds classification uncertainty to this problem. The PUG explicitly defines native AGB per full pixel and advises users to supply their own forest mask; it does not supply within-pixel biomass allocation.

A mapped-pure-forest intersection is reported descriptively, not used to discard changing or mixed support. The presence union, presence intersection and pure intersection have different selection rules. Using persistent pure cells alone would exclude appearing/disappearing forest strata and could remove the conversions the empirical study seeks to evaluate.

The provider's small-plot validation harmonization multiplies a temporally adjusted forest-plot density by forest fraction to construct a comparable reference quantity (PVIR v7, p. 18; PVP v6, p. 15). This introduces assumptions about reference representativeness and nonforest contributions; it does not observe the unknown component densities of a mixed map cell. It is therefore distinct from multiplying an already full-cell-normalized map density by forest fraction. Likewise, model-derived maximum-biomass layers used to constrain retrieval are not independently validated upper bounds on true forest-component density within a mixed cell (ATBD v7, pp. 73–79). Neither operation alone narrows the minimal allocation sets.

S8.2 Reference information and dependence

Collection 11's general ATBD reports about 75,000 error-assessment samples after excluding 10,000 Amazon training samples from an 85,152-point frame. Its detailed validation section

describes interpreted years through 2024, whereas the collection extends through 2025. The inspected public download page labels reference datasets Collection 9. Neither a collection-wide overall accuracy nor a nominal reference-point label establishes a change-error model for these Collection 11 endpoints. A reference audit must verify compatible interpretation labels, independent training/validation roles, inclusion probabilities, design weights and the target transition classes.

The Cerrado appendix documents GEDI-derived canopy-height information in training-mask refinement. CCI's structural inputs have distinct roles: ICESat GLAS supports canopy-density–height calibration, while ICESat-2 heights and national or subnational inventory statistics support the current height–biomass relation, supplemented by FRA statistics where inventory information was not identified. The ATBD explicitly excludes GEDI Level-4 biomass from ingestion and states that the validation inventory-plot database was not used to calibrate or fine-tune this relation (ATBD v7, pp. 57–59, 67–71). Consequently, neither a generic LiDAR label nor the word 'inventory' establishes reuse of the same observations. Structural comparisons, historical inputs and current calibration must be distinguished. Exact auxiliary overlap and its implications for error dependence between the selected products remain to be traced. Classification errors, temporal dependence and spatial covariance still require appropriate reference evidence; different publishers are not a basis for assuming independent errors.

S9 Reference-artifact coverage and compatibility

Three public reference representations linked from the MapBiomas reference-point page are inspected: CSV and vector archives and a complete XLSX worksheet. The page labels the downloads Collection 9 and describes interpretation through 2024. The CSV and XLSX share annual field content, whereas the vector artifact differs in temporal fields and label representation. File path years and modification dates are not evidence of interpretation epoch or verified Collection 11 reference identity. Exact archive/member hashes and field inventories are supplied with the derived audit.

Table S9: Reference-artifact endpoint coverage. Counts are records with nonblank entries, not validated observations, weighted area estimates or accuracy measurements. The CSV and XLSX 2023 entries each include 11 explicitly labelled *NÃO OBSERVADO*.

Property	CSV archive	Vector archive	XLSX workbook
Records	85,152	85,152	85,152
Annual class-field range	1985–2023	1985–2022	1985–2023
2015 nonblank entries	85,152	85,152	85,152
2023 nonblank entries	4,053	No field	4,053
Points inside the fixed footprint	7	7	7
Local paired 2015/2023 entries	0	0	0

The complete workbook has one worksheet, 135 fields and 85,152 distinct identifiers. All identifiers are shared with the CSV. Across all records, the 117 common annual `CLASS`, `COUNT` and `BORDA` field payloads match after trimming whitespace, normalizing Unicode to NFC and representing integral numerals consistently; no semantic class crosswalk is imposed. Both 2015 and 2023 class tokens, both training-role fields and longitude/latitude coordinates also agree, with coordinate comparison tolerance 10^{-8} degrees. The worksheet has no formula cells. These are source-representation checks, not map accuracy or an independent validation sample. The workbook repeats the CSV's endpoint incompleteness and does not supply additional local paired reference labels.

The CSV contains textual annual classes, including explicit non-observation labels. Its absent 2023 entries number 81,099. The vector's numeric 2015 class field includes legacy codes that

require a source-specific legend. No Collection 11 code-to-name or forest-land-use equivalence is presumed, and no mapped-class confusion matrix or transition accuracy is computed.

All 85,152 identifiers are shared by the two artifacts. CSV longitude/latitude coordinates agree with vector point geometries to within 10^{-8} degrees. The vector's `LON` and `LAT` attribute values are reversed relative to the geometry at every record: the first attribute corresponds to latitude and the second to longitude. Positions are therefore taken from the source's WGS84 point geometry, not those attribute names. This is a verified within-artifact coordinate inconsistency, not a positional validation against independent ground control.

In the CSV, `AMOSTRAS=Treinamento` and `AMOSTRA_AM=1` each identify 9,999 records; the remaining 75,153 have a blank training-text entry and indicator zero. The general ATBD describes exclusion of about 10,000 Amazon training points from assessment. The actual flag count and its relation to the documented assessment population require reconciliation. The vector artifact lacks those role fields. Unflagged rows and distinct interpreter agreement do not establish independence from mapping training or previous collections.

Both artifacts contain probability/weight fields, but their design semantics and frame correspondence are not certified here. Decimal-comma values in the CSV are parsed explicitly. The vector has 19 records with zero probability and zero weight, whereas the CSV has positive numeric entries throughout. Source/design reconciliation is needed before probability weighting; these fields are not ecological observations. Even the common 2015 interpreter-count and border fields differ at 2,161 and 1,011 shared identifiers, respectively. These are cross-artifact payload differences, not estimates of map accuracy. The two public representations are not independent reference replications.

The seven locations in the local footprint provide no recorded 2023 classes in any of the three inspected representations. A compatible reference design with appropriately interpreted paired endpoints and documented inclusion/role information is needed before class-change or stock-change validation. These findings concern the three examined public representations; they are not an exhaustive claim that no suitable reference information exists.

S9.1 AGBref stock-reference compatibility

The AGBref descriptor distinguishes harmonized target-epoch stocks from original field dates. Its preprocessing adjusts references within ten years of a target epoch using IPCC growth rates; its timeliness flag uses a three-year criterion. At 500 m and 1 km, the minimum-plot criterion is more than one plot. Local representativity and a strict combined filter are separate criteria (preprint PDF pp. 7–8). Temporal harmonization does not create a measured repeat census, and uncertainty in that adjustment does not independently validate a satellite change estimate.

The open Zenodo record 20793072, file `agb-ref-v2.zip`, supplies named 2005, 2010, 2015 and 2020 products but no named 2023 product. Complete 2010, 2015 and 2020 500 m CSVs and its unnamed CSV were inspected. Each has 200,152 rows and one released aggregate centroid inside the fixed native rectangle, at 49.6825°W, 14.2175°S. Centroid inclusion is not proof of compatible polygon support or original plot representativeness. The three labelled local records have distinct aggregate identifiers, $N = 1$, AGB zero, and zero minimum-plot, large-size, local-representativity and strict-filter flags. SDs are approximately 22.55, 22.55 and 23.02 Mg/ha. Their timeliness flags are zero, one and one. The unnamed record has no date field and is not assigned a 2023 epoch.

The selected exports do not provide original census dates, source-provider codes or original plot identity. Consequently, neither the repeated centroid nor the epoch labels establish an independent repeated-plot pair. The observed plot count also fails the stated 500 m minimum criterion, independently of the zero AGB estimate. No local biomass comparison is admitted from these rows. This conclusion concerns the inspected support and primary interval, not all inventories or every coarser reference product. A four-row, attributed derived subset and the eligibility fields are supplied with the data; no full reference archive is redistributed.

S10 Conditional allocation sets on mapped domains

For a native cell of area A_i and fixed published full-cell density $\widehat{d}_{it}$, write $q_{it} = A_i\widehat{d}_{it}$. Consider an operational mapped group g with area fraction f_{it}^g. Conditional on exact cover, density and epoch correspondence, let its unknown biomass allocation be s_{it}^g. Under nonnegativity alone,

$$s_{it}^g = 0 \quad (f_{it}^g = 0), \qquad s_{it}^g = q_{it} \quad (f_{it}^g = 1), \qquad s_{it}^g \in [0, q_{it}] \quad (0 < f_{it}^g < 1). \tag{S9}$$

For a mixed cell and any $s \in [0, q]$, choosing component means $s/(Af)$ and $(q-s)/(A(1-f))$ attains the allocation with nonnegative densities. No finite component-density cap, minimum biomass threshold, growth constraint or cross-cell coupling is included. Consequently these are exact allocation bounds within the specified minimal algebraic model, not a claim that every extreme is ecologically plausible.

Let P_t^g be the map-pure cells and H_t^g all map-present cells. Summing the independently compatible allocations gives

$$\widehat{\mathcal{I}}_t^g = \left[\sum_{i \in P_t^g} q_{it},\ \sum_{i \in H_t^g} q_{it} \right] = [L_t^g, U_t^g]. \tag{S10}$$

All intermediate aggregate values are compatible in this model. With no additional temporal constraint, the endpoint difference set is

$$\widehat{\mathcal{D}}^g = [L_{2023}^g - U_{2015}^g,\ U_{2023}^g - L_{2015}^g]. \tag{S11}$$

Subtracting the lower endpoints or the upper endpoints alone would not give the full compatible difference set. The interval is conditional on published point maps, not a frequentist confidence set, posterior interval or guaranteed enclosure of true biomass. AGB-SD values are checked for valid support but are not converted to change uncertainty without an appropriate joint error model.

S10.1 Topological support and numerical safeguards

The classifications use the Collection 11 subsets described in S8, and the native biomass grids and indices match the observation-support rectangle in S7. Both density and AGB-SD are valid at all 65,536 target cells at each endpoint. The observed zero counts remain 44,546 and 44,767, respectively. Zero density is not recoded as nonforest, and missing cells are not substituted with zeros.

For each axis, an integer incidence matrix records every finer interval having strictly positive overlap with a native interval. Products of the two axis-incidence matrices with the finer binary group labels give integer counts c_{it}^g of intersecting group pixels; corresponding total counts are $n_i > 0$. Absent, pure and mixed cases are determined by $c = 0$, $c = n$ and $0 < c < n$, respectively, rather than by a rounded forest-fraction threshold. This is an exact topology classification for the recorded grids, not a certification of the ecological classes. Area-weighted fractions recalculated on those overlaps agree with the stored cover fractions to within 3.34×10^{-16}.

Each full-pixel density is multiplied by its WGS84 cell area, not a nominal 100 m × 100 m area or a forest fraction. Pure cells number 791 and 876 for formations, and 6,056 and 5,616 for formations plus savanna. Mixed and changing cells are not discarded. The full 61,830.3436 ha rectangle remains the domain, and the pure-cell intersection across years is not used as the analysis sample.

All difference bounds were calculated from unrounded endpoint bounds; displayed endpoints and differences are rounded independently.

The two formations allocation sets are $[44.2303, 279.1932] \times 10^3$ Mg and $[55.2216, 326.2323] \times 10^3$ Mg. The corresponding difference set is $[-223.9717, 282.0020] \times 10^3$ Mg. Including savannas gives $[229.7548, 434.4732] \times 10^3$ Mg and $[256.3634, 489.2885] \times 10^3$ Mg, with difference

set $[-178.1098, 259.5338] \times 10^3$ Mg. Both sets contain zero. They describe distinct mapped domains; neither validates actual forest land use, a national biomass stock or an account–asset divergence. Additional ecological constraints could narrow an allocation model, whereas measurement/classification error requires a separate uncertainty treatment and need not be represented by these endpoints.

The complete XLSX reference representation examined in S9 repeats the CSV's recorded annual reference fields and supplies no additional paired local endpoints. Its equivalence on the compared fields does not certify independent validation.

S11 Matched-support identification experiments

S11.1 Observations, fixed cases and computational scale

The point-map experiments use only the unchanged Brazilian rectangle, 2015 and 2023 native AGB, WGS84 cell areas, Collection 11 area fractions and integer overlap certificates described above. The footprint contains 65,536 cells and 61,830.3436 ha. All cells, including observed zeros, mixed cells and changing labels, are retained. The spatial experiment compares sides $s \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$ native cells. It holds the complete fine-grid group geometry fixed while replacing observed cell masses by their sums within aligned blocks. Thus it isolates loss of mass-allocation information rather than a change in case selection or mapped-group definition. It is not a model of the sensor's point-spread function, an effective-resolution estimate or a new source observation. The analyses are exploratory; the footprint and endpoints were fixed by the earlier geometry-only support rule, not selected using these results.

The full-cell masses are 463,922.52584046 and 529,164.8923703252 Mg. The native-cell positive and negative point-map differences total 142,213.3577547465 and 76,970.9912248813 Mg. Their net is 65,242.3665298653 Mg and their cancellation fraction is 0.7023402135. These values are not adjusted for observation error. The same net is preserved at all nine scales, up to floating-point summation. At side 64, 128 and 256, the blockwise negative sum and cancellation fraction are zero. No conclusion about true fine-scale change follows without a compatible error model.

S11.2 Sharp block bounds and nested information sets

Within block b, the observed mass is q_b and the mapped group either has zero area, occupies the whole block, or has positive area together with a positive-area complement. Under nonnegative unconstrained density, the group mass is respectively 0, q_b, or any value in $[0, q_b]$. The mixed case is attained by assigning component means x/A_g and $(q_b - x)/A_h$ for arbitrary $x \in [0, q_b]$, where $A_g, A_h > 0$. Summing these independent intervals yields the block endpoint bounds in the main text; taking the cross-endpoint extrema yields sharp difference bounds. The union's topology is certified absent only if every constituent count is zero, and pure only if every group count equals its total count. Counts, not rounded area fractions, determine these categories.

Every assignment satisfying finer block totals satisfies the coarser totals obtained by summation. Hence the projected group-mass sets can only expand under nested coarsening. In this experiment the group and complement each occupy positive area at both endpoints, giving the one-window difference set $[-Q_{2015}, Q_{2023}]$. Its width is 993,087.4182107851 Mg. The native-cell widths are 505,973.6696403596 and 437,643.5603800343 Mg, reductions of 49.0504% and 55.9310%, respectively (rounded values in the main table). Width is an identification metric conditional on fixed point maps; its reduction is not empirical accuracy or confidence-coverage improvement.

For cancellation, the triangle inequality gives $\sum_b |\sum_{i \in b} \Delta q_i| \leq \sum_i |\Delta q_i|$, while the net is unchanged. At a nonzero net, $K = 1 - |\sum_i \Delta q_i| / \sum_i |\Delta q_i|$ therefore cannot increase under nested

aggregation. If the total variation is zero, K is undefined rather than assigned zero. The present nonzero-net data do not encounter that undefined case.

S11.3 Constructive endpoint witnesses

Let c_{it}^g and n_i be group and total intersection counts. Define $x_{it}^{\mathrm{lo}} = q_{it}\mathbf{1}\{c_{it}^g = n_i\}$ and $x_{it}^{\mathrm{hi}} = q_{it}\mathbf{1}\{c_{it}^g > 0\}$. The minimum-change witness uses $(x_{i,2015}^{\mathrm{hi}}, x_{i,2023}^{\mathrm{lo}})$ and the maximum uses $(x_{i,2015}^{\mathrm{lo}}, x_{i,2023}^{\mathrm{hi}})$. For each endpoint and cell, group and complement masses are nonnegative, sum to the unchanged observed cell mass, and respect absent/pure topology. The component densities in mixed cells are formed on their positive mapped areas, so the witness retains the exact mapped fraction rather than changing the map. No finite density cap, temporal growth restriction or ecological plausibility claim is added. Convex interpolation of the endpoint allocations attains all intermediate differences. The witnesses are constructed separately for each overlapping mapped group; simultaneous attainment of both groups' extreme bounds is not asserted.

The four aggregate witness certificates report 65,536 constraint-satisfying cells at each endpoint. The minimum and maximum changes reproduce the unrounded primary difference bounds to numerical summation precision. Native-cell masses and allocations are generated in memory for these checks but are not redistributed. Aggregate certificates are supplied as `data/extremal_witness_certificates_F11.csv`; the complete nested-scale results are in `data/resolution_frontier_F11.csv` and `data/cancellation_by_scale_F11.csv`.

S11.4 Sharp joint set for nested cover groups

Let A denote formations, S added savanna and O the complement, so $H = A + S$ denotes formations plus savanna. Their integer overlap counts are c^A, $c^S = c^H - c^A$ and $c^O = n - c^H$. The nesting $0 \le c^A \le c^H \le n$ is checked for every cell. Given cell mass q, its feasible (Q^A, Q^H) allocation is the convex hull of whichever of (q, q), $(0, q)$ and $(0, 0)$ correspond to components with positive mapped area. Cells with one available component give a point, two give an edge, and three give a triangle. Component area fractions are unchanged; the nonnegative component densities are free.

The endpoint polygon is the Minkowski sum of these cell polygons, and the difference polygon is the Minkowski sum of the 2023 endpoint polygon and the negative 2015 polygon. Every cell edge is horizontal, vertical or parallel to $(1, 1)$. The support function of a Minkowski sum is the sum of support functions, so its possible facet orientations are contained in the same three directions, including their opposite normals. Its exact half-space description is therefore given by the lower and upper support bounds for a, h and $h - a$. Independent cellwise endpoint optimization gives precisely the three univariate difference intervals in Eq. S12:

$$\mathcal{D}_{\mathrm{joint}} = \{(a, h) : a \in \widehat{\mathcal{D}}^A,\ h \in \widehat{\mathcal{D}}^H,\ h - a \in \widehat{\mathcal{D}}^S\}. \tag{S12}$$

This establishes a sharp joint set, not merely necessary marginal constraints. It also explains why simultaneously choosing arbitrary marginal extrema need not be feasible. No cross-group statistical independence is assumed.

For computation, cells are aggregated into the seven nonempty availability patterns of the three components, retaining each pattern's total mass at each endpoint. Let M_{pt} be that mass and z_{pkt} the mass assigned to an available component k. The constraints are $z_{pkt} \ge 0$, $\sum_k z_{pkt} = M_{pt}$ and $z_{pkt} = 0$ for an absent component. This reduction is exact: for $M_{pt} > 0$, assign each constituent cell's mass in the proportions z_{pkt}/M_{pt}; a zero-mass pattern requires only zero allocation. All cell totals, absent components and pure cells are then preserved. Thus the optimization does not discard native-cell mass constraints.

Four linear programs maximize a common signed margin $\epsilon \ge 0$ under $\sigma_A a \ge \epsilon$ and $\sigma_H h \ge \epsilon$, for the four pairs $(\sigma_A, \sigma_H) \in \{-1, 1\}^2$. All attain strictly positive margins. The coherent witness

differences (a, h), in 10^3 Mg, are $(-223.97, -178.11)$, $(-119.38, 119.38)$, $(103.68, -103.68)$ and $(282.00, 259.53)$. Reconstructed cell allocations preserve mass to at most 1.43×10^{-14} Mg per cell. This tolerance concerns numerical summation, not ecological uncertainty. The calculated joint polygon has six vertices. A separate support-function optimization over the pattern constraints agrees with the half-space polygon over 73 evaluated directions to 1.17×10^{-10} Mg; the sharpness result rests on the geometric argument above, not that finite check.

Pattern summaries, quadrant certificates and polygon vertices are supplied as `data/joint_availability_patterns_F11.csv`, `data/joint_quadrant_certificates_F11.csv` and `data/joint_polygon_vertices_F11.csv`. They disclose aggregate derived quantities, not native raster arrays. The common ratio sensitivity in the next subsection is evaluated separately for each mapped group and is not asserted to define one jointly calibrated three-component ecological model.

S11.5 Sharp bounds under an assumed density contrast

In a mixed cell write $d = f d_g + (1-f) d_h$, where $0 < f < 1$ and $q = Ad$. The restrictions $d_g \le R d_h$ and $d_h \le R d_g$ with $R \ge 1$ imply

$$\frac{f}{f + R(1-f)} \le \frac{s_g}{q} \le \frac{Rf}{Rf + 1 - f}, \qquad q > 0.$$

The lower endpoint is attained with $d_h = R d_g$, and the upper with $d_g = R d_h$. Both endpoints respect the observed mixture. If $q = 0$, nonnegativity gives $s_g = 0$ without division. Pure/absent cells are fixed separately and impose no comparison with an unobserved zero-area component. These formulas use area fractions, not intersection-count fractions. Integer topology supplies exact zero/one values for certified absent/pure cells; floating area-integration deviations of at most 3.34×10^{-16} at these certified endpoints are set to their exact values. Every mixed-cell fraction is retained strictly inside $(0, 1)$, without a tolerance-based membership decision.

Independent summation and cross-endpoint subtraction yield sharp aggregate difference bounds for each R. They expand monotonically with R, reduce to homogeneous allocation at $R = 1$, and converge to the primary unrestricted bounds as $R \to \infty$. The common bound applies to both endpoints and every mixed cell; it is an additional structural assumption, not a density parameter fitted from the maps. The strictly positive homogeneous differences lose strict sign identification when the lower bound reaches zero, at $R = 1.4555078889$ for formations and $R = 1.9495126065$ for formations plus savanna. A bracketed root calculation in log-R space locates these boundaries. They are conditional sensitivity boundaries, not estimated ecological contrasts, confidence thresholds or universal necessary conditions under every possible auxiliary-information model.

Table S10: Conditional density-contrast sensitivity, in 10^3 Mg. R is assumed, not estimated or calibrated. $R = 1$ is homogeneous allocation; $R = \infty$ is the primary unrestricted model.

Mapped domain	R	Difference lower	Difference upper
Formations	1	28.7755	28.7755
Formations	1.5	-2.2882	59.8019
Formations	2	-23.8622	81.3096
Formations	3	-52.9802	110.2998
Formations	5	-86.5922	143.7427
Formations	10	-125.3424	182.3383
Formations	100	-197.9409	255.0793
Formations	∞	-223.9717	282.0020

Mapped domain	R	Difference lower	Difference upper
Formations + savanna	1	41.8699	41.8699
Formations + savanna	1.5	16.2717	67.4158
Formations + savanna	2	-1.5677	85.1567
Formations + savanna	3	-25.7930	109.1672
Formations + savanna	5	-54.0850	137.0918
Formations + savanna	10	-87.3635	169.7909
Formations + savanna	100	-152.9575	234.0673
Formations + savanna	∞	-178.1098	259.5338

The sparse contrasts and boundary results are supplied in `data/density_contrast_sensitivity_F11.csv` and `data/density_contrast_sign_boundaries_F11.csv`; a dense $1 \leq R \leq 10$ evaluation supports the figure in `data/density_contrast_curve_F11.csv`. No finite contrast is adopted for the primary result. Even a justified allocation restriction would leave map/classification error, temporal correspondence and independent change validation to be addressed separately.

Table S11: Same-footprint information comparison under the unrestricted allocation model. Widths are in 10^3 Mg. Reduction compares native-cell mass observations with the one-window mass observation, retaining identical mapped domains and point maps. Neither information set identifies a group-change sign.

Mapped domain	One-window width	Native-cell width	Reduction (%)
Formations	993.09	505.97	49.05
Formations + savanna	993.09	437.64	55.93

S12 National trajectory and source-support specifications

Actual national trajectories use all 183 eligible identifiers in 2015–2023; the separate 2010–2020 account comparison retains 182 eligible members of that cohort. Account contributions are $100\Delta B_k/B_{\text{total},0}$ percentage points. Forest-component decomposition uses the symmetric product identity for source account area and the implied gha/ha factor. That factor belongs to the accounting lineage and is not independent forest productivity or stock evidence. The account-domain and component contribution tables retain all observed signs.

National MapBiomas tables are summed at non-overlapping leaf classes within each published geography. Brazilian formations are $\{3, 5, 6, 49\}$, with $\{4, 7\}$ added only for the distinct formations-plus-savanna analysis. Indonesian forest is $\{3, 5, 76\}$, oil palm is $\{35\}$ and the broader agricultural target is $\{9, 21, 35, 40\}$. Congolese source forest is $\{3, 4, 5\}$, dense forest plus mangrove is $\{3, 5\}$ and plantation remains $\{9\}$. These are not interchangeable forest definitions. All source footprints, zero/no-data conventions and class crosswalks remain in the data.

The Brazilian reported footprint is 850,644,939.994 ha and the Indonesian Collection 4.1 footprint is 189,044,390.245 ha. Consecutive transition margins reproduce finest-geography extent totals to less than 0.000005 and 0.0000003 ha, respectively. Those residuals measure arithmetic consistency, not map accuracy. Indonesia's Collection 3 footprint exceeds Collection 4.1 by 43,130.051 ha. A difference of total areas is not a geometry symmetric difference, and source/support differences cannot be isolated as a pure collection effect.

The Congo archive folder is named "wide", but its CSV members have long schema with explicit year, class and area fields. The country member contains 234 rows, nine observed leaf classes and all 26 years, 2000–2025. Units are established by the version-pinned source script's `pixelArea().divide(10000)`, not inferred from an unlabelled `area` header. Source statistics

round class areas to two decimal digits. The country label footprint ranges from 231,404,518.04 to 231,404,518.08 ha across years. Biome and province exports do not sum to this footprint; they remain separate support summaries. Neither near-constant label area nor net footprint differences certify a complete, disjoint geometry partition or the extent of unobserved forest.

Congo's country forest-group endpoint difference is $-3,303,015.650$ ha (-2.014673%). The symmetric endpoint net-area break-even requirement is $1,651,507.825$ ha per endpoint, not an uncertainty estimate. Its ATBD describes a 60-km corridor on each side of the Miombo–Mosaic boundary, additional classification trained on temporally stable map labels outside the corridor, and replacement restricted to selected vegetation classes. It also documents class confusion and irregular temporal series. These procedures support the provenance description, not independent paired change validation. No stock loss or forest-to-agriculture conversion is inferred from the marginal area changes.

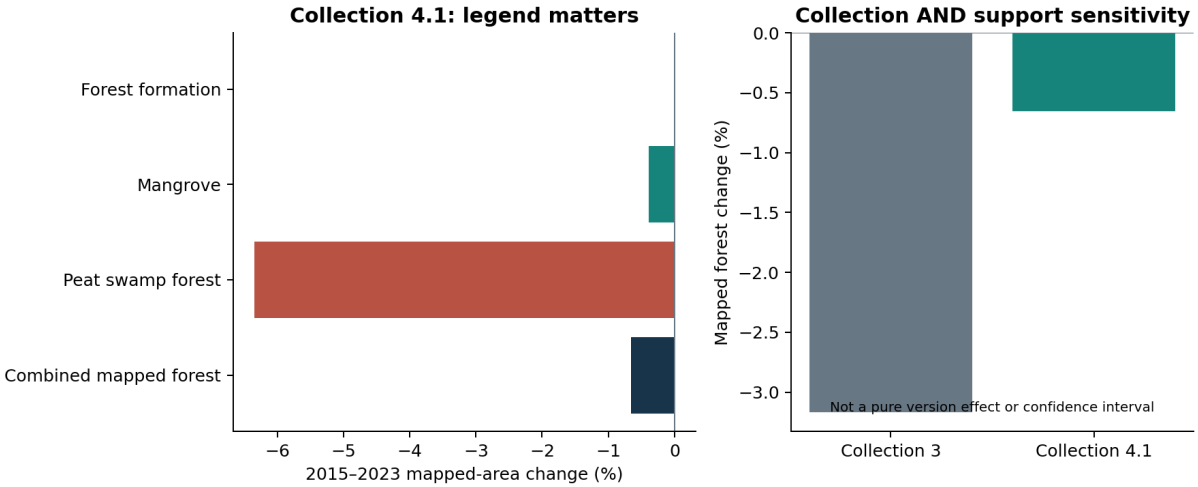


Figure S2: Indonesia definition and collection/support sensitivity, 2015–2023. Left: changes in source forest subclasses and their union. Right: C3 versus C4.1 mapped forest changes; differing support prevents interpretation as a pure collection effect or confidence interval. Main national results summarize this sensitivity. Source: MapBiomas Indonesia; original numerical figure retained.

Table S12: Source-specific class groups and geographic support checks. Numeric class codes are initiative-specific. Footprint differences are not geometry symmetric differences.

Source	Mapped group	Support qualification
Brazil C11	3, 5, 6, 49; plus 4, 7 only for F+S	850.645 million ha; annual margins reconciled.
Indonesia C4.1	3, 5, 76; oil palm 35 separate	189.044 million ha; C3 footprint 43,130.051 ha larger.
DR Congo C1	3, 4, 5; plantation 9 separate	Country footprint 231.405 million ha; province/biome totals not forced into a partition.

S13 Sharp endpoint-cohort bounds and independent numerical certificates

Let G be a layered directed acyclic graph with vertices (t, i) and edge capacities $T_{tij} \geq 0$. For a marked cohort whose complete origin is the specified 2015 group, the feasible set is

$$\mathcal{X} = \{x : 0 \leq x \leq T,\ x_{2015,ij} = T_{2015,ij}\mathbf{1}\{i \in F\},\ Ax = 0\},$$

where A imposes conservation at every interior vertex. For a destination D, the objective is $c^T x = \sum_i \sum_{j \in D} x_{2022,ij}$. The minimum and maximum give unique endpoint-cohort bounds, not event totals.

If capacities are exactly balanced, the unmarked flow $T - x$ is also nonnegative and conserved. Both flows decompose into paths; their continuum of area parcels constitutes a compatible land history. Conversely, any compatible history generates a feasible marked flow. Thus the bounds are sharp for continuous-area aggregate constraints, not necessarily for an integer original-pixel lattice or ecologically restricted histories. Any path may traverse intermediate classes or repeat conversion. Restricting flows to reported fixed geographic units removes cross-unit reallocations. Its projected feasible set is a subset of the national aggregate set, so raw extrema cannot weaken. Numerical outward allowances may make rounded endpoints marginally wider and are reported separately.

The seven group–target cases comprise four Brazilian and three Indonesian cases. Two optimization directions at national level and within each of 49 or 38 units yield 634 programmes. Hectares are scaled by 10^5 inside the solver. The recorded feasibility and gap limit is 0.02 ha per programme, including L^1 conservation, source imbalance and bound-violation diagnostics. The independent witness check reconstructs constraints and verifies primal feasibility, dual stationarity, complementarity and primal–dual objective agreement from the deposited capacities without invoking the optimizer. Source-balanced ideal-model sharpness and finite-precision numerical certificates are distinct claims.

The finest-geography endpoint bounds in million ha are:

Table S13: Exact fine-geography endpoint-cohort ranges (million ha), corresponding to main Figure 5. Values include outward numerical allowances; they are feasible mapped ranges, not confidence intervals.

Country	Initial group	2023 target	Lower	Upper
BRA	formations	nonforest	11.203746901	22.710751994
BRA	formations	cropland	0.000000000	7.014430883
BRA	formations plus savanna	nonforest	18.373845177	35.847255651
BRA	formations plus savanna	cropland	0.009385753	8.661678113
IDN	mapped forest	nonforest	1.312001539	4.424121169
IDN	mapped forest	oil palm	0.111612129	2.325782972
IDN	mapped forest	agriculture	0.126800213	3.763620519

For true endpoint sets F, D and mapped sets $\widehat{F}, \widehat{D}$, a union-set argument gives

$$\left|\mu(F \cap D) - \mu(\widehat{F} \cap \widehat{D})\right| \leq \mu(F \triangle \widehat{F}) + \mu(D \triangle \widehat{D}).$$

Consequently a mapped lower bound L transfers only as $\max(0, L - E)$ for a compatible combined set-error budget E. Indonesia's oil-palm lower bound requires $E < 111,612.129$ ha; an equal split would be approximately 55,806 ha per endpoint. Neither an overall classification score nor net endpoint area error establishes this requirement.

Table S14: Finnish inventory source cells and reported volume standard errors. Units are million m^3 after conversion from the source thousand-m^3 cells; source identities are pinned in the numerical dependency.

Inventory	Volume	SE	Sheet	Cells
NFI12	2475.192	13.001	20a	N61/O61
NFI13	2551.964	13.534	Table 55	N60/O60

S14 Inventory source cells, period operators and dependence bounds

Finnish original source cells are NFI12 supplementary `20a!N61/O61`, 2,475,192 and 13,001 thousand m^3, and NFI13 `Table55!N60/O60`, 2,551,964 and 13,534 thousand m^3. The values after the slash are sampling standard errors, not 95% halfwidths. Both inventories include Åland. The complete source table identifies units and inventory cycles. The approximate normal Bonferroni difference band, including source rounding, is $[17.293137220, 136.250862780]$ million m^3. A worst-covariance second-moment sensitivity gives $[-41.896127566, 195.440127566]$ million m^3, conditional on unbiased errors with the stated error variances. Both exclude unbounded differential bias.

The Finnish biomass API reports 1,744 and 1,786 Mt for total living trees including belowground components. Independently rounded stems plus branches/foliage sum to 1,350 and 1,384 Mt. No biomass standard error is inferred from volume error.

French 2013–2017 volume and 95% halfwidth are 2,716 and 45 million m^3 in the France row of *Mémento 2018*, printed pp. 28–29, PDF page 16 (a two-page spread). The row is the national France row, not Île-de-France. The associated methodology, PDF pp. 5, 9, 16–17, describes five-campaign pooling, protocol changes, production-forest categories including poplar, the diameter threshold and unusable-wood deduction. Current workbook national totals are read from `BV Propriété`, row `Total`, in the respective Volume XLSX member:

Table S15: French published national volume estimates and 95% marginal halfwidths. Campaign windows are pooled observations, not exact annual endpoints.

Campaign window	Estimate (million m^3)	Published 95% halfwidth
2013–2017	2,716	45
2018–2022	2,811	46
2019–2023	2,827	46
2021–2025	2,916	47

French workbook sheet dimensions incorrectly advertise `A1:A1`; parsing resets those dimensions before iterating the actual seven columns. The readme specifies 95% intervals, production forest and the stock definition. The latest includes 30,136 inventory points, 29,502 outside poplar plantations and 634 in poplar plantations. Changed physical-accessibility and planted-forest breakdowns are not compared across old/new classifications. Matching nominal stock definitions does not certify exact retrospective cubage or unusable-wood harmonization. A residual differential-method bias budget is left explicit rather than set to zero.

With full-unit allowance $r = 1$ million m^3 for each estimate and each halfwidth, the per-contrast 90% band is

$$\Delta\widehat{V} \pm [h_0 + h_1 + 4r].$$

For an approximate 95% marginal-normal band, use

$$\Delta\widehat{V} \pm \left[\frac{z_{1-0.05/(2m)}}{z_{0.975}}(h_0 + h_1 + 2r) + 2r\right],$$

with $m = 2$ for one contrast or $m = 4$ for simultaneous coverage of the four endpoints and hence all three latest-cycle contrasts. No independence is assumed. The longest difference is +200 million m^3: the per-contrast 90% band is $[104, 296]$, the pairwise approximate 95% band is $[90.502180, 309.497820]$, and the four-endpoint approximate 95% band is $[78.209883, 321.790117]$. All three earlier-to-latest contrasts remain in the deposited table, including the shorter unresolved ones. A systematic differential bias with absolute bound B expands either side by B; no measured bound is asserted. These are conditional inventory-volume results, not calibrated national 2015–2023 AGB.

S15 Canadian biomass change and uncertainty representations

The Canadian source is NFI version 3.1, November 2025. Tables 17.0 and 17C.0 report stock and annualized change by forest type and age class; accompanying `sep` and `intv` files report relative standard errors and interval halfwidths. CSV values retain greater numerical precision than their HTML renderings. The HTML footers establish combined remeasured 2007–2017 and projected 2006–2017 data, removal of strong artificial-change indicators, and omitted Arctic ecozones plus the Nunavut portions of Taiga Plains and Hudson Plains. The omitted area is 262,189.89 thousand ha, not assumed to carry zero forest biomass.

The stock total is 30,525.199 Mt with labelled RSE 0.70% and halfwidth 418.4005 Mt, approximately consistent with the usual large-sample interval relation after RSE rounding. Annualized total change is +17.639 Mt/year with labelled RSE 1.16% and halfwidth 38.9180 Mt/year. The latter ratio $38.918/(17.639 \times 0.0116)$ is 190.204, not approximately 2. Individual change rows also fail the ordinary relation, so a rounded percentage alone cannot supply a change standard error. The point, interval and RSE HTML tables all carry the same version 3.1, November 2025 label; a simple labelled-version mismatch does not resolve the discrepancy. The published halfwidth is retained as labelled and its confidence level is not independently certified for this export. The general estimation document uses default Type-I probability 0.05 and a national critical value near 2, but it is not a current implementation specification for these projected change tables.

Table S16: Canadian annualized change by contemporaneous forest type, with provider-labelled halfwidths. No confidence level is assigned here; components are not fixed biological cohorts.

Source forest type	Annualized change (Mt/year)	Labelled halfwidth
Coniferous	−16.158	34.1130
Mixedwood	−16.115	12.2560
Broadleaf	+43.065	13.9775
Non-treed	+6.847	2.3140
Total	+17.639	38.9180

All four component rows and the total are retained; none is selected as a confirmatory sign test. They refer to stocks classified by contemporaneous forest type, not fixed biological cohorts. Source totals reconcile to rounding. The direct total interval $[-21.279, 56.557]$ Mt/year includes zero, and full-CSV-unit outward rounding gives $[-21.2801, 56.5581]$. Projection, classification error and artificial-change filtering remain separate from sampling precision. The source design permits useful inventory-domain evidence without imposing an exact paired census, but it does not support transfer to omitted geography or the primary endpoints.

S16 Reference compatibility protocols

The Brazilian Collection 11 accuracy appendix uses reference collection 3 for 1985–2024, with boundary-point exclusion, interpreter weighting, sparse-class exclusions and map-dependent legend harmonization. The examined public collection-2.1 archive is an older reference represen-

tation. Of 85,152 points, 4,053 have nonblank 2015/2023 pairs; 4,041 remain after removing non-observed labels and the archived training role; 3,042 are also nonedge at both endpoints. These counts do not certify a compatible national probability-design sample. Local zero paired-reference eligibility remains a property of the original footprint and examined artifacts, not a global absence claim. The different current/old training counts are preserved as provenance differences, not reconciled by assumption. A newly retrieved vector archive currently linked from the official download page has an October 2026 URL but 85,152 records and 126 fields ending in 2022; no 2023 or 2024 annual fields occur. Its recent URL therefore does not establish the current reference collection or fill the endpoint gap. The DBF header is checked without expanding the full table.

Indonesia's Collection 4.1 ATBD describes independent annual references through 2022 and does not supply a current global Collection 4.1 accuracy estimate in its collection summary. Older pooled accuracy does not calibrate its 2015–2023 change. Congo's ATBD describes limitations and future validation improvements but does not establish a compatible endpoint change interval. None of these findings proves that an adequate reference design is impossible. Compatible marginal reference intervals, direct inventory-change estimates, defensible covariance bounds or bounded bias can be sufficient; exact paired field censuses and known covariance are not universal prerequisites.

Table S17: All French earlier-to-2021–2025 volume contrasts (million m^3), including full-unit rounding. Normal rows are approximate and conditional on marginal normality. An unbounded differential-method bias is not covered.

Initial cycle	Construction	Difference	Lower	Upper
2013–2017	90% union	200	104.000000	296.000000
2013–2017	95% normal, 2 ends	200	90.502180	309.497820
2013–2017	95% normal, 4 ends	200	78.209883	321.790117
2018–2022	90% union	105	8.000000	202.000000
2018–2022	95% normal, 2 ends	105	-5.641414	215.641414
2018–2022	95% normal, 4 ends	105	-18.064480	228.064480
2019–2023	90% union	89	-8.000000	186.000000
2019–2023	95% normal, 2 ends	89	-21.641414	199.641414
2019–2023	95% normal, 4 ends	89	-34.064480	212.064480

Source attribution and availability

National extensions: MapBiomas Brazil Collection 11; MapBiomas Indonesia Collection 4.1 and Collection 3; MapBiomas DR Congo Collection 1, statistics, legend and ATBD; IGN national standard volume tables and 2018 historical national row; Canada's NFI Tables 17.0 and 17C.0 and associated interval/RSE tables; Finnish Luke APIs and the supplementary tables of Korhonen et al. (2021, 2024). Source URLs and SHA-256 identities accompany derived outputs. MapBiomas DR Congo data/code access and attribution are documented by its ATBD, p. 19. IGN factual numerical extractions are attributed to IGN; provider PDF bodies and raw workbooks are not redistributed. The numerical workbook preserves original table meanings and adds separate source and scope notes.

National accounts: Lo, K., et al. (2026), National Ecological Footprint and Biocapacity Accounts, Edition 2026, results and metadata v1.0; https://footprint.info.yorku.ca/data/. Derived account tables and figures are distributed under CC BY-SA 4.0.

Biomass: Santoro, M., and Cartus, O. (2026), ESA CCI Biomass v7.0; https://doi.org/10.5285/6429d1aafe1e43b9b414e4a5a7f8b903. The ESA Climate Change Initiative, Biomass CCI project and data providers are acknowledged. Product definitions follow the Product User Guide v7.0, 19 March 2026; https://climate.esa.int/en/projects/biomass/key-documents/.

Boundary source licences and OSM attribution are retained in the data package. OSM-derived boundary data and associated support databases are supplied under ODbL 1.0; other source terms are identified separately. The native-window arrays, biomass guide and provider implementation are not redistributed.

MapBiomas Project, Collection 11 of the Annual Land Cover and Land Use Maps of Brazil, accessed 29 October 2026: https://brasil.mapbiomas.org/downloads/. Local cover fractions are derived by the stated class grouping and WGS84 area intersections. Public reference-artifact links are listed at https://brasil.mapbiomas.org/downloads/pontos-de-validacao/; the audit supplies factual schemas and derived aggregate completeness checks, not the source archives. Public-data attribution conditions are stated at https://brasil.mapbiomas.org/faq/?tema=dados. Raw country land-cover rasters and source method documents are not redistributed.

AGBref: Araza, A., Bielig, C., and Flores, R. A. A. (2026), *AGBref—Global Reference Dataset for Above-Ground Biomass (AGB)*, Zenodo record 20793072; https://doi.org/10.5281/zenodo.20793072, CC BY 4.0. The limited local subset is derived from the released aggregate CSVs, with source member and record attribution. Descriptor: Araza, A., et al. (2026), Research Square preprint; https://doi.org/10.21203/rs.3.rs-8211898/v1. Source files and reference metadata do not establish paired field observations.

Source hashes, metadata, derived account-availability tables and computational methods are available at https://github.com/MIKEAA2020/general-sustainability/tree/main/batch%209/agent%202/ecomod_empirical_followup/F11_A01. Raw data, provider technical reports and provider code are not redistributed.

Integrated manuscript, national tables, new analyses, source identities and dependency verification: https://github.com/MIKEAA2020/general-sustainability/tree/main/batch%209/agent%202/ecomod_empirical_followup/F14_A01.

S17 Provider technical references

Document-specific page references above use PDF page numbers. The cited editions are:

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